

Euclid's Elements — Book XI

Proposition 18

Formal Reconstruction — Technical Documentation

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CGJTEAMLAB

If a straight line is at right angles to any plane, then all the planes through it are also at right angles to the same plane.

1 Introduction

Proposition XI.18 is the point at which the formal reconstruction must make a deliberate choice about the meaning of perpendicular planes.

A modern textbook may define two planes to be perpendicular when one of them contains a line perpendicular to the other. That definition is mathematically natural in Euclidean space, but it is unsuitable for a source-faithful reconstruction of Euclid XI.18. Under that definition the proposition would be nearly true by definition: the given plane through the normal line would already contain a normal to the reference plane.

The present reconstruction instead follows Euclid XI.Def.4. A plane ρ is perpendicular to a plane π along their common section s when every line of ρ perpendicular to s is perpendicular to π . XI.18 must therefore prove a universal statement about all such lines in ρ .

The production file is

```
CGJteamLab/Wyler/Proposition11_18_wyler.lean
```

with the main theorem

```
euclid_proposition_11_18_wyler
```

The supporting perpendicular-plane definition is isolated in

```
CGJteamLab/Wyler/HilbertWylerPlanePerpendicular.lean
```

and the neutral planar bridge used in the proof is isolated in

```
CGJteamLab/Wyler/HilbertWylerPerpendicularParallel.lean
```

2 Euclid's Statement

Euclid states Proposition XI.18 as follows:

If a straight line is at right angles to any plane, then all the planes through it are also at right angles to the same plane.

Let l be perpendicular to the plane π at O , and let ρ be any plane containing l . The hypothesis is

$$l \perp \pi, \quad l \subset \rho.$$

The conclusion is

$$\rho \perp \pi.$$

In the formal reconstruction this conclusion means perpendicularity in the sense of XI.Def.4. It is therefore stronger than the bare existential statement that ρ happens to contain one normal to π .

3 Classical Configuration

The classical configuration contains:

- the reference plane π ;
- a line l perpendicular to π at O ;
- an arbitrary plane ρ through l ;
- the common section

$$s = \rho \cap \pi.$$

Since $O \in l \subset \rho$ and $O \in \pi$, the point O lies on the section s .

To verify XI.Def.4 one takes an arbitrary point $F \in s$ and an arbitrary line $g \subset \rho$ perpendicular to s at F . The goal is

$$g \perp \pi.$$

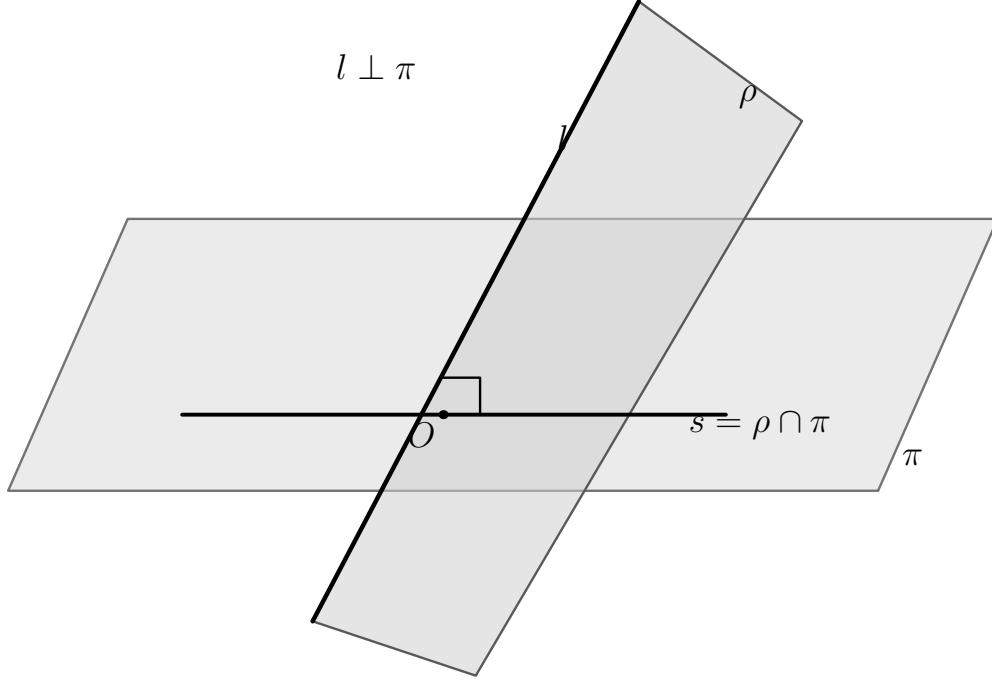


Figure 1: The classical configuration of Proposition XI.18. The line l is perpendicular to the reference plane π at O , and the arbitrary plane ρ contains l . Its common section with π is $s = \rho \cap \pi$. Proposition XI.18 proves that ρ is perpendicular to π in the sense of XI.Def.4.

4 Euclid's Classical Proof

Euclid begins with a straight line AB perpendicular to the reference plane and lets an arbitrary plane DE pass through AB . The line CE is the intersection of DE with the reference plane.

A point F is chosen arbitrarily on CE , and inside the plane DE a line FG is drawn perpendicular to CE .

Because AB is perpendicular to the reference plane, it is perpendicular to every line of that plane through its foot; in particular it is perpendicular to CE . Thus AB and FG are two lines in the same plane DE , both perpendicular to CE .

Euclid then uses I.28 to infer

$$AB \parallel FG.$$

By XI.8, since $AB \perp \pi$ and $FG \parallel AB$,

$$FG \perp \pi.$$

Finally XI.Def.4 says precisely that if the lines drawn in the plane DE perpendicular to the common section CE are perpendicular to the reference plane, then DE is perpendicular to the reference plane.

The arbitrary choice of DE gives the proposition for every plane through the original normal line.

5 Classical Dependency Pattern

The classical proof has the following dependency pattern:

$$\begin{array}{c}
l \perp \pi, \quad l \subset \rho \\
\Downarrow \\
s = \rho \cap \pi \\
\Downarrow \\
l \perp s \\
\Downarrow \\
\text{choose } F \in s, \quad g \subset \rho, \quad g \perp s \\
\Downarrow \\
l \parallel g \\
\Downarrow \\
\text{XI.8 : } \quad g \perp \pi \\
\Downarrow \\
\text{XI.Def.4 : } \quad \rho \perp \pi.
\end{array}$$

There is one formal refinement hidden by the traditional diagram: if $F = O$, the two perpendiculars l and g need not be parallel as distinct lines; they are the same line. The formal bridge must therefore return

$$l = g \quad \text{or} \quad l \parallel g.$$

6 What Is Genuinely Three-Dimensional Here?

XI.18 mixes a planar argument with a genuinely spatial conclusion.

The genuinely spatial data are:

- the normal relation $l \perp \pi$;
- containment $l \subset \rho$;
- the exact plane meet $s = \rho \cap \pi$;
- the transport theorem XI.8, which moves line–plane perpendicularity along a spatially parallel line.

The step comparing l and an arbitrary g is planar. Both lines lie in the single plane ρ , and both are perpendicular to the same line s . That local argument belongs naturally to the induced geometry $\text{PlaneGeo}(\rho)$.

Thus XI.18 has a clean two-level structure:

ambient 3D meet and normality + planar perpendicular comparison + ambient XI.8 transport.

7 Formal Statement

The production theorem is:

```

theorem euclid_proposition_11_18_wyler
  [H : HilbertIncidence Geo]
  [HilbertPlaneIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  [HSI : HilbertSpaceIncidence Geo]
  [_HSO : HilbertSpaceOrder
    (Geo := Geo) (H := H) (S := S)]
  [HSC : HilbertSpaceCongruence
    (Geo := Geo) (H := H) (S := S)]
  [HSE : HilbertSpaceEuclidean Geo]
  (pi rho : S.Plane)
  (l : Geo.Line)
  (O : Geo.Point)
  (hPerp :
    HilbertLinePerpendicularPlaneAt Geo l pi O)
  (hlrho :
    HilbertLineInPlane Geo l rho) :
  HilbertSpacePlanesPerpendicular Geo rho pi

```

The conclusion is the directed predicate

```
HilbertSpacePlanesPerpendicular Geo rho pi
```

rather than a symmetric primitive relation.

This directionality is intentional: ρ is the plane in which lines perpendicular to the common section are tested, while π is the plane to which those lines must be perpendicular.

8 Hilbert–Wyler Architecture

The proof uses three layers which should not be conflated.

8.1 Faithful XI.Def.4 plane perpendicularity

The lower predicate is

```
HilbertSpacePlanesPerpendicularAlong Geo rho pi s
```

and records four facts:

- $s \subset \rho$;
- $s \subset \pi$;
- s is exactly the common point set of ρ and π ;

- every line $g \subset \rho$ perpendicular to s at F is perpendicular to π at F .

The public predicate

```
HilbertSpacePlanesPerpendicular Geo rho pi
```

adds $\rho \neq \pi$ and existentially packages the section line s .

This definition deliberately avoids the modern shortcut

ρ is perpendicular to π iff ρ contains a line perpendicular to π .

Using that shortcut would remove the mathematical content of XI.18.

8.2 Neutral same-perpendicular bridge

The principal helper is

```
hilbert_coplanar_perpendiculars_to_same_line_eq_or_parallel_wyler
```

It states that inside one ambient plane, if l and g are both perpendicular to the same line s , then

$$l = g \quad \text{or} \quad l \parallel g.$$

No Euclidean parallel axiom is used in this helper.

The proof splits according to the feet O, F .

If $O = F$, uniqueness of the perpendicular direction in $\text{PlaneGeo}(\rho)$ gives $l = g$.

If $O \neq F$, a hypothetical intersection point K of l and g would create a nondegenerate triangle OFK with two right angles. The neutral I.17 machinery rules this out, so the lines are disjoint; since both lie in ρ , they are spatially parallel.

8.3 XI.8 transport

The only genuinely Euclidean step in the final proposition is the use of

```
euclid_proposition_11_8
```

which transports perpendicularity to a plane along a parallel line:

$$l \parallel g, \quad l \perp \pi \quad \implies \quad g \perp \pi.$$

This is where the production theorem requires `HilbertSpaceEuclidean Geo`.

9 Ambient Space and PlaneGeo

The global proposition is formulated in the ambient geometry:

```
pi rho : S.Plane
l s g   : Geo.Line
O F     : Geo.Point
```

The exact section

$$s = \rho \cap \pi$$

is obtained by the ambient plane-intersection API.

The induced geometry

```
PlaneGeo Geo rho
```

appears only inside the neutral helper comparing l and g . There it is used to turn ambient line perpendicularities into ordinary planar right angles and to invoke planar uniqueness/nonintersection arguments.

This separation reflects the mathematics accurately:

- plane meets and line–plane perpendicularity are spatial;
- two lines perpendicular to the same line inside one carrier plane are a planar problem.

10 Lean Proof Architecture

The production proof follows the source closely but makes the hidden branch structure explicit.

10.1 Stage 1: recover the exact common section

From

$$l \perp \pi \text{ at } O$$

the incidence data give

$$O \in l, \quad O \in \pi.$$

Since $l \subset \rho$,

$$O \in \rho.$$

The proof first shows

$$\rho \neq \pi.$$

Indeed, if $\rho = \pi$, then $l \subset \pi$, contradicting the fact that a line perpendicular to a plane cannot itself lie in that plane.

The exact plane intersection theorem then gives a line s satisfying

$$O \in s, \quad s \subset \rho, \quad s \subset \pi,$$

and for every point X ,

$$X \in \rho \cap \pi \iff X \in s.$$

Thus s is not merely an auxiliary common line; it is the exact common section required by XI.Def.4.

10.2 Stage 2: the given normal is perpendicular to the section

Since $s \subset \pi$ and $O \in s$, the line–plane normality hypothesis immediately gives

$$l \perp s \text{ at } O.$$

10.3 Stage 3: verify the universal XI.Def.4 condition

Take arbitrary

$$g \subset \rho, \quad F \in g \cap s, \quad g \perp s \text{ at } F.$$

The neutral coplanar helper gives

$$l = g \quad \text{or} \quad l \parallel g.$$

In the equality branch the original normality certificate is reused. The formal proof still checks that the test foot F equals the original foot O , using uniqueness of the foot of a perpendicular line to a plane.

In the parallel branch XI.8 gives a point D such that

$$g \perp \pi \text{ at } D.$$

But $F \in s \subset \pi$ and $F \in g$, so foot uniqueness gives

$$F = D.$$

Hence

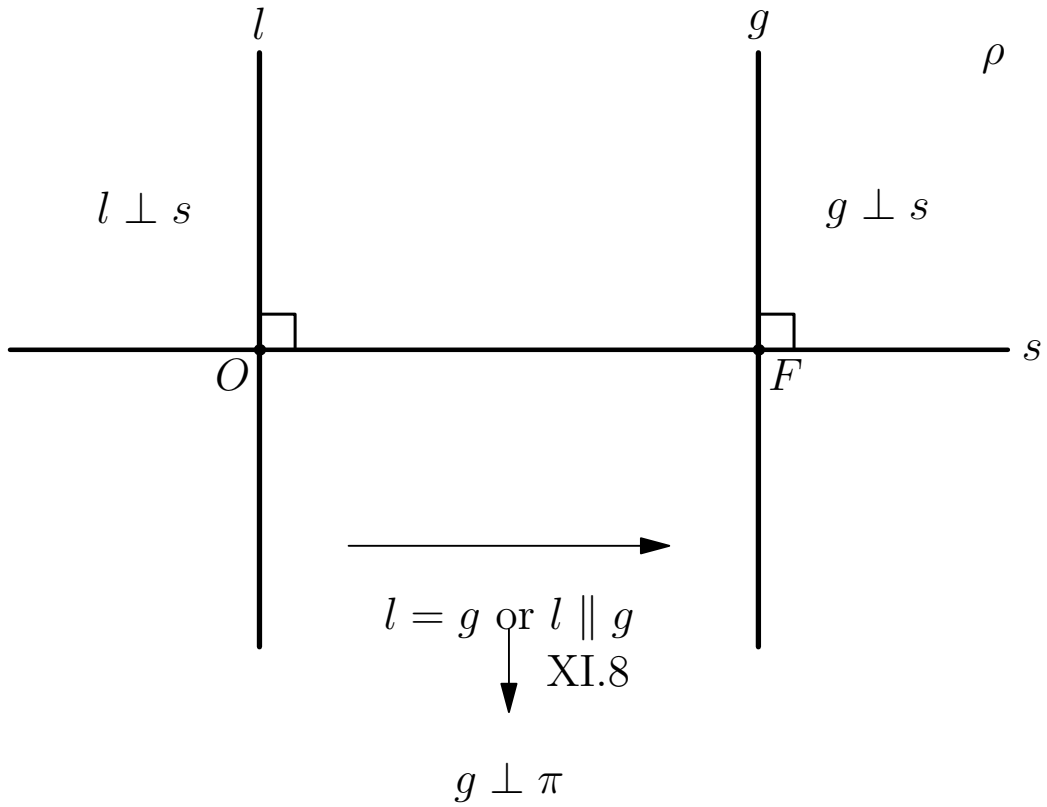


Figure 2: The formal core of XI.18 inside the plane ρ . The given normal l and an arbitrary test line g are both perpendicular to the common section s . If their feet coincide then $l = g$; if the feet are distinct then the neutral planar argument gives $l \parallel g$. In the parallel branch XI.8 transports $l \perp \pi$ to $g \perp \pi$.

$$g \perp \pi \quad \text{at } F.$$

Since g and F were arbitrary, XI.Def.4 is satisfied.

11 Wyler-Style Flat Reconstruction

The Wyler-style contribution to XI.18 is the exact treatment of the common section and the separation of flat incidence from metric information.

The key flat identity is

$$\text{carrier}(s) = \text{carrier}(\rho) \cap \text{carrier}(\pi).$$

The proof does not define perpendicular planes in flat language. Instead it uses the flat meet to identify the carrier on which the metric condition of XI.Def.4 is tested.

The architecture is therefore

$$\begin{aligned} & \text{Meet}(\rho, \pi) = s \\ & \Downarrow \\ & \text{metric fact } l \perp s \\ & \Downarrow \\ & \text{for arbitrary } g \subset \rho, \ g \perp s \\ & \Downarrow \\ & g = l \text{ or } g \parallel l \\ & \Downarrow \\ & g \perp \pi. \end{aligned}$$

This is characteristic of the Wyler refactor: flat operations determine the configuration, while congruence/right-angle structure supplies the metric content.

12 Hidden Formal Data

Several facts hidden in the classical figure become explicit in Lean.

12.1 The two planes must be distinct

The target predicate requires

$$\rho \neq \pi.$$

This is proved rather than assumed. Equality would place the given normal line l inside the plane to which it is perpendicular.

12.2 The common section is exact

Euclid says that CE is the intersection of the two planes. The formal proof records the stronger universal certificate

```
forall X : Geo.Point,
  (S.OnPlane X rho /\ S.OnPlane X pi) <=> H.OnLine X s
```

which is exactly what later propositions such as XI.19 need.

12.3 The test line is arbitrary

XI.Def.4 is universal. The proof must handle an arbitrary pair

```
g : Geo.Line
F : Geo.Point
```

with $g \subset \rho$ and $g \perp s$ at F .

This is why proving only the existence of one normal in ρ would be insufficient.

12.4 Parallelism has an equality branch

Euclid writes that two lines perpendicular to the same line are parallel. In the formal setting the feet may coincide. Thus the correct statement is

$$l = g \vee l \parallel g.$$

This branch distinction is mathematically real, not Lean bureaucracy.

12.5 The transported foot must be identified

XI.8 returns a perpendicular line with an existentially supplied foot D . XI.Def.4 requires perpendicularity specifically at the chosen section point F . The proof therefore uses foot uniqueness to prove $F = D$.

13 Relationship with Earlier and Later Book XI Propositions

XI.18 is strongly connected to the preceding metric development.

XI.3 / plane-intersection infrastructure. The proof needs the exact common section of ρ and π .

XI.4 metric core. The lower helper for coplanar perpendiculars uses the right-angle machinery developed around XI.4 after passing to $\text{PlaneGeo}(\rho)$.

XI.8. This is the decisive Euclidean transport step: a line parallel to a normal to a plane is itself normal to that plane.

XI.14 neutral right-angle helper. The distinct-foot branch uses the neutral impossibility of a nondegenerate triangle with two right angles, ultimately reducing to Euclid I.17.

XI.19. The next proposition uses the XI.Def.4 infrastructure introduced for XI.18. In particular, it extracts normal lines from perpendicular-plane certificates and combines them with XI.13.

Thus XI.18 is not only an isolated theorem. It establishes the correct interface for plane perpendicularity used by XI.19 and later spatial metric arguments.

14 Relationship with Hilbert's Axiom Groups

The dependency structure is informative.

The neutral helper

```
hilbert_coplanar_perpendiculars_to_same_line_eq_or_parallel_wyler
```

uses:

- Group I incidence;
- Group II order through the planar triangle machinery;
- Group III congruence/right-angle structure.

It does not use the Euclidean parallel axiom.

The full production theorem additionally assumes

```
HilbertSpaceEuclidean Geo
```

because the parallel branch invokes XI.8.

Thus Group IV enters XI.18 at one sharply localized place:

$$l \parallel g, \quad l \perp \pi \quad \implies \quad g \perp \pi.$$

The definition of perpendicular planes, the exact meet, and the same-perpendicular neutral bridge do not themselves require Group IV.

15 Classical DAG versus Formal DAG

The comparison is especially useful because the formal proof exposes the equality branch and the exact meaning of XI.Def.4.

Euclid.

$$\begin{array}{c}
l \perp \pi, \quad l \subset \rho \\
\Downarrow \\
s = \rho \cap \pi \\
\Downarrow \\
l \perp s \\
\Downarrow \\
g \subset \rho, \quad g \perp s \\
\Downarrow \\
l \parallel g \\
\Downarrow \\
\text{XI.8 : } g \perp \pi \\
\Downarrow \\
\text{XI.Def.4 : } \rho \perp \pi.
\end{array}$$

Lean / Hilbert–Wyler.

$$\begin{array}{c}
l \perp \pi \text{ at } O, \quad l \subset \rho \\
\Downarrow \\
O \in \rho \cap \pi, \quad \rho \neq \pi \\
\Downarrow \\
s = \text{Meet}(\rho, \pi) \\
\Downarrow \\
l \perp s \text{ at } O \\
\Downarrow \\
\forall g, F : g \subset \rho, \quad g \perp s \text{ at } F \\
\Downarrow \\
l = g \vee l \parallel g \\
\Downarrow \\
\begin{cases} l = g : & F = O, \text{ reuse } l \perp \pi, \\ l \parallel g : & \text{XI.8} \Rightarrow g \perp \pi \text{ at } D, \quad F = D \end{cases} \\
\Downarrow \\
\text{XI.Def.4} \\
\Downarrow \\
\rho \perp \pi.
\end{array}$$

The additional formal nodes do not change Euclid’s argument. They make precise the carrier intersection, the universal quantifier in XI.Def.4, and the degenerate equality case suppressed by the diagram.

16 Structural Lesson

XI.18 is a useful test of whether the formalization preserves theorem content rather than merely reproducing modern equivalent definitions.

If plane perpendicularity were defined by

$$\rho \perp \pi \quad :\Longleftrightarrow \quad \exists l \subset \rho, \quad l \perp \pi,$$

then the proposition would collapse immediately from the hypothesis that ρ contains the given normal l .

The source-faithful XI.Def.4 predicate avoids that collapse. It asks for a universal behavior of all lines in ρ perpendicular to the common section. XI.18 then becomes a genuine theorem establishing the equivalence between the existence of a plane normal and Euclid's common-section criterion, in the direction needed here.

This is also an architectural lesson for the Wyler program. Exact flat meets should describe where the metric condition is tested, but they should not absorb the metric condition itself.

17 Conclusion

Proposition XI.18 has the compact classical form

$$l \perp \pi, \quad l \subset \rho \quad \Longrightarrow \quad \rho \perp \pi.$$

The formal reconstruction reveals three distinct ingredients behind this statement.

First, the planes ρ and π have an exact common section

$$s = \rho \cap \pi.$$

Second, inside ρ , the given normal l and every XI.Def.4 test line g are both perpendicular to s . Neutral planar geometry therefore gives the precise alternative

$$l = g \quad \text{or} \quad l \parallel g.$$

Third, in the parallel branch XI.8 transports the original normality of l to g , and foot uniqueness places that normality at the required point of the common section.

The result is a source-faithful reconstruction in which XI.18 remains a genuine theorem rather than a consequence of a modernized definition. It also supplies exactly the perpendicular-plane interface used in the subsequent proof of XI.19.